Assignment

System Engineering and Project Management

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Question 1: House Building Project

The Project Network
House Building Project Data

Number	Activity	Predecessor	Duration
1	Design house and obtain financing	--	3 months
2	Lay foundation	1	2 months
3	Order and receive materials	1	1 month
4	Build house	2,3	3 months
5	Select paint	2, 3	1 month
6	Select carpenter	5	1 month
7	Finish work	4, 6	1 month

Using the above table find the following:

1. All possible paths
2. Critical path
3. ES, EF, LS, LF
4. Slack

Solution 1:

Step 1: All Possible Paths

From the table:

- 1 → 2 → 4 → 7
- 1 → 3 → 4 → 7
- 1 → 2 → 5 → 6 → 7
- 1 → 3 → 5 → 6 → 7

Step 2: Duration of Each Path

- Path 1: 1-2-4-7 = 3 + 2 + 3 + 1 = **9 months**
- Path 2: 1-3-4-7 = 3 + 1 + 3 + 1 = **8 months**
- Path 3: 1-2-5-6-7 = 3 + 2 + 1 + 1 + 1 = **8 months**
- Path 4: 1-3-5-6-7 = 3 + 1 + 1 + 1 + 1 = **7 months**

Step 3: Critical Path

Critical path = longest path

Critical Path = 1⇒ 2 ⇒ 4 ⇒ 7

Project Duration = 9 months

Step 4: ES, EF, LS, LF

(Forward + Backward pass)

Activity	ES	EF	LS	LF
1	0	3	0	3
2	3	5	3	5
3	3	4	4	5
4	5	8	5	8
5	5	6	6	7
6	6	7	7	8

Activity	ES	EF	LS	LF
7	8	9	8	9

Step 5: Slack

Slack = LS - ES

Activity	Slack
1	0
2	0
3	1
4	0
5	1
6	1
7	0

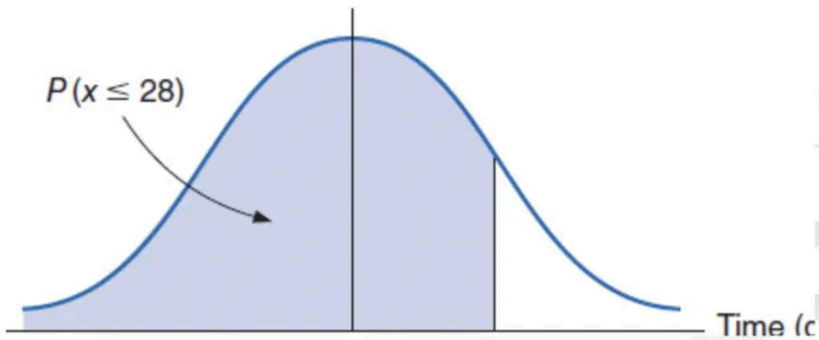
Activities with **0 slack** → **Critical activities (1,2,4,7)**

Question 2: PERT Analysis

Step 4: Determine the Probability That the Project Will be Completed in 28 days or less ($\mu = 24$, $\sigma = \sqrt{5}$)

$$Z = (x - \mu)/\sigma = (28 - 24)/\sqrt{5} = 1.79$$

Corresponding probability from Table A.1, Appendix A, is .4633 and $P(x \le 28) = .4633 + .5 = .9633$.



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Using the last example, Find the probability that the project will be completed in

- ⬛ 24 days or less.
- ⬛ 20 days or less.
- ⬛ 32 days or less.
- ⬛ Briefly explain your observations

Solution 2:

We assume project completion time follows **normal distribution**.

Formula:

Z = (x − mu) / Sigma

Sigma = Root(5) = 2.236

mu = 24

1. Probability for 24 days or less

$Z = (24 - 24) / 2.236 = 0$

From Z-table:

$P(Z \le 0) =$ **0.5**

ans: 50%

2. Probability for 20 days or less

$Z = (20 - 24) / 2.236 = -1.79$

From Z-table:

Area = 0.4633

Since Z is negative:

$P = 0.5 - 0.4633 =$ **0.0367**

ans: 3.67%

3. Probability for 32 days or less

$Z = (32 - 24) / 2.236 = 3.58$

From Z-table:

$P \approx$ **0.9998**

ans: 99.98%

Observation

- At mean (24 days) → probability = 50%
- Much earlier (20 days) → very low probability
- Much later (32 days) → almost certain completion
